

Test of Mathematics for University Admission, 2020 Paper 1

Worked Solutions

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Introduction for students

These solutions are designed to support you as you prepare to take the Test of Mathematics for University Admission. They are intended to help you understand how to answer the questions, and therefore you are strongly encouraged to **attempt the questions first** before looking at these worked solutions. For this reason, each solution starts on a new page, so that you can avoid looking ahead.

The solutions contain much more detail and explanation than you would need to write in the test itself – after all, the test is multiple choice, so no written solutions are needed, and you may be very fluent at some of the steps spelled out here. Nevertheless, doing too much in your head might lead to making unnecessary mistakes, so a healthy balance is a good target!

There may be alternative ways to correctly answer these questions; these are not meant to be ‘definitive’ solutions.

The questions themselves are available on the ‘**Preparing for the test**’ section on the Admissions Testing website.

Question 1

We start by writing the expression as a sum of powers of x :

$$\frac{x^3 - 5x^2}{2x\sqrt{x}} = \frac{x^3 - 5x^2}{2x^{\frac{3}{2}}} = \frac{x^{\frac{3}{2}} - 5x^{\frac{1}{2}}}{2} = \frac{1}{2}x^{\frac{3}{2}} - \frac{5}{2}x^{\frac{1}{2}}$$

We can now differentiate this expression to get

$$\frac{1}{2} \times \frac{3}{2}x^{\frac{1}{2}} - \frac{5}{2} \times \frac{1}{2}x^{-\frac{1}{2}}$$

We now need to convert this into the form given in the options, so we write it as

$$\frac{3}{4}\sqrt{x} - \frac{5}{4\sqrt{x}} = \frac{3\sqrt{x}}{4} - \frac{5}{4\sqrt{x}} = \frac{3x}{4\sqrt{x}} - \frac{5}{4\sqrt{x}} = \frac{3x - 5}{4\sqrt{x}}$$

which is option C.

Question 2

We can use the factor theorem here. If we write $f(x) = 2x^3 + px^2 + q$, then the question tells us that $f(-\frac{1}{2}) = 0$ and $f(2) = 0$. Substituting these in gives

$$\begin{aligned} -\frac{1}{4} + \frac{1}{4}p + q &= 0 \\ 16 + 4p + q &= 0 \end{aligned}$$

We can subtract the first equation from the second to cancel the qs , giving

$$\frac{65}{4} + \frac{15}{4}p = 0$$

so $p = -\frac{65}{15} = -\frac{13}{3}$. Therefore $q = -16 - 4p = -\frac{48}{3} + \frac{52}{3} = \frac{4}{3}$.

It follows that $2p + q = -\frac{26}{3} + \frac{4}{3} = -\frac{22}{3}$, which is option C.

Question 3

We first work on each inequality separately. Each one is a product of factors, so we consider when each factor is positive or negative.

For the first inequality, $(x + 4)(x + 3)(1 - x) > 0$, we have the following table:

	$(x + 4)$	$(x + 3)$	$(1 - x)$	product
$x < -4$	−	−	+	+
$-4 < x < -3$	+	−	+	−
$-3 < x < 1$	+	+	+	+
$x > 1$	+	+	−	−

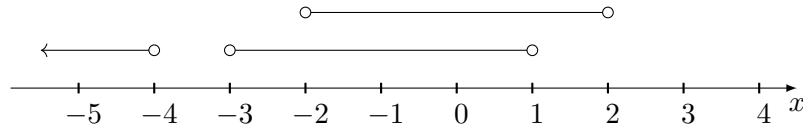
Thus the first inequality holds when $x < -4$ and when $-3 < x < 1$.

Likewise for the second inequality, $(x + 2)(x - 2) < 0$:

	$(x + 2)$	$(x - 2)$	product
$x < -2$	−	−	+
$-2 < x < 2$	+	−	−
$x > 2$	+	+	+

Therefore the second inequality holds when $-2 < x < 2$.

We can combine these by illustrating them on a number line:



Combining these two conditions, they both hold when $-2 < x < 1$, which is option B.

Question 4

Let the arithmetic progression have first term a and common difference d . Let the geometric progression have first term a (as the first term of the geometric progression is the same as the first term of the arithmetic progression) and common ratio r .

Then the first three terms of the geometric progression are a , ar and ar^2 , while the 1st, 4th and 6th terms of the arithmetic progression are a , $a + 3d$ and $a + 5d$.

Therefore $ar = a + 3d$ and $ar^2 = a + 5d$.

The next piece of information we are given is that the sum to infinity of the geometric progression is 12, so $\frac{a}{1-r} = 12$, hence $a = 12(1-r)$.

So we have three equations:

$$\begin{aligned} ar &= a + 3d \\ ar^2 &= a + 5d \\ a &= 12(1-r) \end{aligned}$$

There are several ways to work out the value of a (as required for this question) from these equations; here is one approach.

We can eliminate d from the first two equations by finding 5 times the first equation minus 3 times the second equation; this gives

$$5ar - 3ar^2 = 2a$$

so dividing by a yields $5r - 3r^2 = 2$, giving $3r^2 - 5r + 2 = 0$. This factorises as $(3r-2)(r-1) = 0$, so $r = 1$ or $r = \frac{2}{3}$. Since the geometric progression has a sum to infinity, we must have $|r| < 1$, so $r = \frac{2}{3}$.

Then the final equation gives $a = 4$, which is option D.

Question 5

The line of symmetry can be determined by completing the square of the right hand side of the given equation for S :

$$\begin{aligned} px^2 + 6x - q &= p\left(x^2 + \frac{6}{p}x\right) - q \\ &= p\left(\left(x + \frac{3}{p}\right)^2 - \frac{9}{p^2}\right) - q \\ &= p\left(x + \frac{3}{p}\right)^2 - \frac{9}{p} - q \end{aligned}$$

so the line of symmetry is at $x = -\frac{3}{p}$. Therefore $-\frac{3}{p} = -\frac{1}{4}$, giving $p = 12$.

The curve touches the x -axis at exactly one point, so we must also have $-\frac{9}{p} - q = 0$, giving $q = -\frac{9}{p} = -\frac{9}{12} = -\frac{3}{4}$.

Therefore $p + 8q = 12 - 6 = 6$, which is option A.

Question 6

The denominator is a quadratic in $u = 5^x$, so we can write it as

$$\frac{1}{u^2 - 4u + 7} = \frac{1}{(u - 2)^2 + 3}$$

The maximum value of this fraction is obtained when the denominator takes its minimum value, which is when $u = 2$, giving a value of $\frac{1}{3}$. We must also check that this value can be obtained; this is the case as $5^x = 2$ has a real solution.

Therefore the correct answer is option C.

Question 7

We write these two equations in powers of 2, giving:

$$\begin{aligned}2^{3x} &= 2^{3y+9} \\ 2^{2x+2} &= \frac{2^{4y+4}}{2^{3y+9}} = 2^{(4y+4)-(3y+9)} = 2^{y-5}\end{aligned}$$

Therefore we have

$$\begin{aligned}3x &= 3y + 9 \\ 2x + 2 &= y - 5\end{aligned}$$

We can divide the first of these equations by 3 for simplicity, giving

$$\begin{aligned}x &= y + 3 \\ 2x + 2 &= y - 5\end{aligned}$$

We can now solve these simultaneous equations; subtracting the first from the second gives $x + 2 = -8$, so $x = -10$ and $y = -13$, so $x + y = -23$, which is option A.

Question 8

We can find the maximum value of $f(x)$ by completing the square or by differentiating.

In both methods, we start by expanding the brackets to write $f(x) = -x^2 + (p-2)x + 2p$.

We can then complete the square:

$$\begin{aligned} f(x) &= -(x^2 - (p-2)x) + 2p \\ &= -\left(x - \frac{1}{2}(p-2)\right)^2 - \frac{1}{4}(p-2)^2 + 2p \\ &= -\left(x - \frac{1}{2}(p-2)\right)^2 + \frac{1}{4}(p-2)^2 + 2p \end{aligned}$$

so the maximum value occurs when $x = \frac{1}{2}(p-2)$ and the value at this point is $\frac{1}{4}(p-2)^2 + 2p$.

Alternatively, we can differentiate to get

$$f'(x) = -2x + (p-2)$$

so the stationary point is at $x = \frac{1}{2}(p-2)$; substituting this value into the function $f(x)$ gives the same maximum value as before.

A third approach is to note that the stationary point of a quadratic lies midway between its two roots, that is midway between p and -2 , so at $x = \frac{1}{2}(p-2)$, and then substitute to find the minimum value.

We now require

$$\frac{1}{4}(p-2)^2 + 2p < 4.$$

Multiplying by 4 and rearranging gives $(p-2)^2 + 8p - 16 < 0$; we next expand the brackets and simplify to give $p^2 + 4p - 12 < 0$, so $(p+6)(p-2) < 0$. This quadratic inequality is satisfied exactly when $-6 < p < 2$, which is option D.

Question 9

As $x^2 - 14x + 9 = (x - \alpha)(x - \beta)$, the roots of the quadratic $x^2 - 14x + 9 = 0$ are α and β .

Approach 1: calculate α and β

The most obvious approach to answering this question is to calculate α and β directly using the quadratic formula:

$$\alpha, \beta = \frac{14 \pm \sqrt{14^2 - 4 \times 1 \times 9}}{2} = \frac{14 \pm 2\sqrt{7^2 - 9}}{2} = 7 \pm \sqrt{7^2 - 9} = 7 \pm \sqrt{40} = 7 \pm 2\sqrt{10}$$

But now we need to find the square root of these expressions. We can guess that it has the form $p + q\sqrt{10}$ for some integers p and q . Squaring this gives

$$(p + q\sqrt{10})^2 = (p^2 + 10q^2) + 2pq\sqrt{10}$$

We cannot make $p^2 + 10q^2 = 7$, but if we take $p = 2$ and $q = \pm 1$, we get $14 \pm 4\sqrt{10}$, which is 2α or 2β . So if we divide by $\sqrt{2}$, taking $p = \sqrt{2}$ and $q = \pm \frac{1}{\sqrt{2}}$, we will obtain $\pm\alpha$ and $\pm\beta$. Therefore

$$\sqrt{\alpha}, \sqrt{\beta} = \pm(\sqrt{2} \pm \sqrt{5}).$$

However, we must take the positive values for α and β , so we must have

$$\sqrt{\alpha}, \sqrt{\beta} = \sqrt{5} \pm \sqrt{2}.$$

Therefore

$$\begin{aligned} (x - \sqrt{\alpha})(x - \sqrt{\beta}) &= x^2 - (\sqrt{\alpha} + \sqrt{\beta})x + \sqrt{\alpha}\sqrt{\beta} \\ &= x^2 - ((\sqrt{5} + \sqrt{2}) + (\sqrt{5} - \sqrt{2}))x + (\sqrt{5} + \sqrt{2})(\sqrt{5} - \sqrt{2}) \\ &= x^2 - 2\sqrt{5}x + 3 \\ &= x^2 - \sqrt{20}x + 3 \end{aligned}$$

which is option C.

Approach 2: use properties of α and β

An alternative way of approaching this question is as follows.

If we expand the two quadratics of interest, we have

$$\begin{aligned} (x - \alpha)(x - \beta) &= x^2 - (\alpha + \beta)x + \alpha\beta = x^2 - 14x + 9 \\ (x - \sqrt{\alpha})(x - \sqrt{\beta}) &= x^2 - (\sqrt{\alpha} + \sqrt{\beta})x + \sqrt{\alpha}\sqrt{\beta} \end{aligned}$$

Thus $\alpha + \beta = 14$ and $\alpha\beta = 9$, and we need to determine $\sqrt{\alpha} + \sqrt{\beta}$ and $\sqrt{\alpha}\sqrt{\beta}$. We can take the square root of $\alpha\beta = 9$ to obtain $\sqrt{\alpha}\sqrt{\beta} = 3$.

Next,

$$(\sqrt{\alpha} + \sqrt{\beta})^2 = \alpha + \beta + 2\sqrt{\alpha}\sqrt{\beta} = 14 + 2 \times 3 = 20$$

using the values we have determined already, and so $\sqrt{\alpha} + \sqrt{\beta} = \sqrt{20}$. Therefore the correct answer is option C, as before.

Question 10

The most straightforward way to answer this question is by performing the transformations to the formula in the standard way. An alternative approach is to sketch the graphs, but then it is not immediately clear how we would determine the resulting formula.

We start with $y = 4x^2$.

1. Translating it right by 3 transforms the curve into $y = 4(x - 3)^2$. Then translating it down by 5 transforms it into $y = 4(x - 3)^2 - 5$.
2. Reflecting it in the x -axis (the horizontal axis!) replaces y by $-y$, or equivalently replaces $y = f(x)$ by $y = -f(x)$, giving $y = -4(x - 3)^2 + 5$.
3. Stretching it parallel to the x -axis with scale factor 2 replaces x by $\frac{x}{2}$, giving $y = -4(\frac{x}{2} - 3)^2 + 5$.

We can now expand the resulting formula, to give $y = -4(\frac{x^2}{4} - 3x + 9) + 5 = -x^2 + 12x - 31$, which is option A.

Question 11

At first sight, it seems that there may not be enough information to answer the question, as we only know two points on the quadratic. But if we stretch the graph parallel to the y -axis, though we modify the quadratic, we will not change q , and the areas of R and S will remain equal. So let us proceed. For simplicity, we will write R and S for the areas of these regions.

The quadratic has equation $y = k(x - 2)(x - q)$. By the above comments, we may as well take $k = 1$; if we leave k in the equation, the areas will each be multiplied by k , and when we equate them, we can then divide by k .

The area of R is

$$\begin{aligned} R &= \int_0^2 (x - 2)(x - q) \, dx \\ &= \int_0^2 x^2 - (q + 2)x + 2q \, dx \\ &= \left[\frac{1}{3}x^3 - \frac{1}{2}(q + 2)x^2 + 2qx \right]_0^2 \\ &= \frac{8}{3} - 2(q + 2) + 4q \\ &= 2q - \frac{4}{3} \end{aligned}$$

The area of S is as follows; note that the integral is negative, so we have to negate the integral to obtain the area.

$$\begin{aligned} S &= - \int_2^q (x - 2)(x - q) \, dx \\ &= - \int_2^q x^2 - (q + 2)x + 2q \, dx \\ &= - \left[\frac{1}{3}x^3 - \frac{1}{2}(q + 2)x^2 + 2qx \right]_2^q \\ &= - \left(\frac{1}{3}q^3 - \frac{1}{2}(q + 2)q^2 + 2q^2 \right) + \left(\frac{8}{3} - 2(q + 2) + 4q \right) \\ &= - \left(-\frac{1}{6}q^3 + q^2 \right) + \left(2q - \frac{4}{3} \right) \\ &= \frac{1}{6}q^3 - q^2 + 2q - \frac{4}{3} \end{aligned}$$

As $R = S$, we have

$$2q - \frac{4}{3} = \frac{1}{6}q^3 - q^2 + 2q - \frac{4}{3}$$

and therefore $\frac{1}{6}q^3 - q^2 = 0$. As $q > 2$, we cannot have $q = 0$, so we can divide by q^2 to give $\frac{1}{6}q - 1 = 0$. It follows that $q = 6$, which is option E.

You may have noticed that the value of the integral at $x = 2$ appeared in both expressions and then cancelled out when we equated them. Another way to think about this is that as the integral from 0 to 2 is positive and the integral from 2 to q is negative, then if the areas R and S are the same, we must have the integral from 0 to q equal to 0. So we could have calculated:

$$\begin{aligned} 0 &= \int_0^q (x - 2)(x - q) \, dx \\ &= \int_0^q x^2 - (q + 2)x + 2q \, dx \\ &= \left[\frac{1}{3}x^3 - \frac{1}{2}(q + 2)x^2 + 2qx \right]_0^q \\ &= -\frac{1}{6}q^3 + q^2 \end{aligned}$$

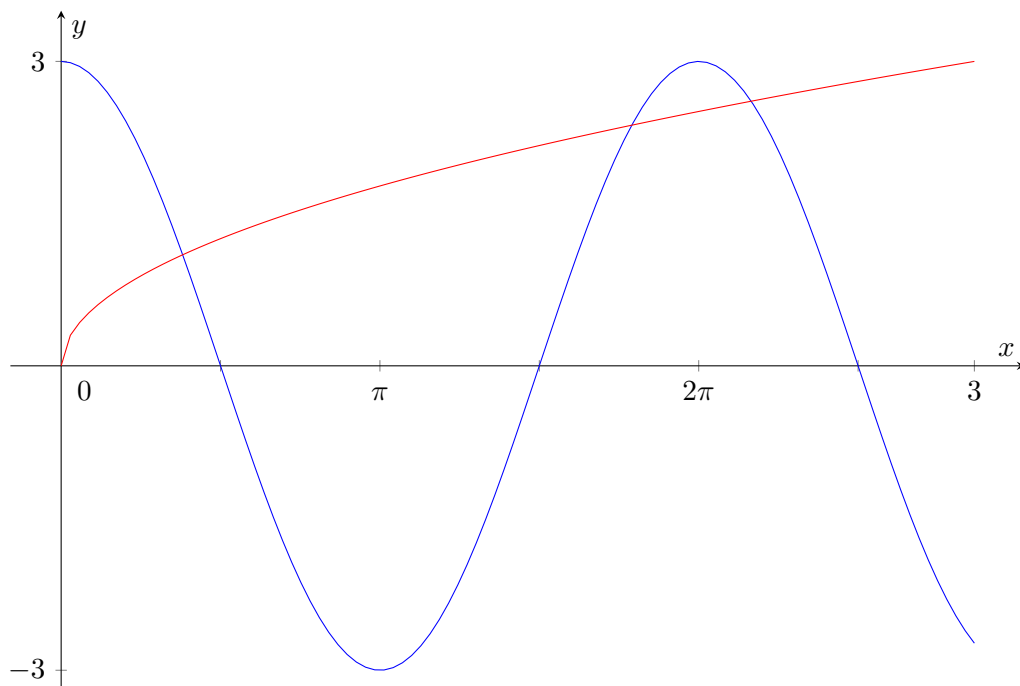
giving $q = 6$ as before.

Question 12

We can make some immediate observations about this equation:

- The left hand side takes values between -3 and $+3$.
- The right hand side is only defined for $x \geq 0$ and is always non-negative, and when $x > 9$, $\sqrt{x} > 3$.

Therefore we need only consider the region $0 \leq x \leq 9$. Note that $3\pi \approx 9.5$, so we have just less than 1.5 periods of the cosine function in this range, and $3\cos 9 < 0$. We can therefore sketch the pair of graphs $y = 3\cos x$ and $y = \sqrt{x}$ and observe how many times they intersect:



We see that there are 3 points of intersection, and hence 3 solutions to the equation, which is option D.

Question 13

We have not learnt a “trinomial theorem” for expanding expressions like $(1+x+y^2)^7$, so instead we treat it as a binomial expression. If we write it as $(1+(x+y^2))^7$, we will then have a series of complicated expressions like $(x+y^2)^3$ that we have to handle. But if we write it as $(x+(1+y^2))^7$, we will be separating the x ’s and the y ’s, which is much simpler for our needs.

We have

$$\begin{aligned}(1+x+y^2)^7 &= (x+(1+y^2))^7 \\ &= x^7 + \binom{7}{1}x^6(1+y^2) + \cdots + \binom{7}{5}x^2(1+y^2)^5 + \cdots\end{aligned}$$

so we need only consider the x^2 term; note also that $\binom{7}{5} = \binom{7}{2} = \frac{7 \times 6}{2} = 21$. So the x^2 term is $21x^2(1+y^2)^5$.

We next want the y^4 term of $(1+y^2)^5$, so we expand this too using the binomial theorem:

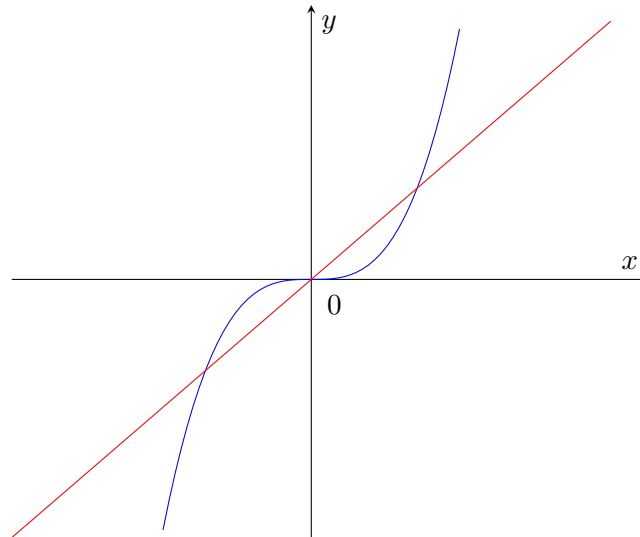
$$(1+y^2)^5 = 1 + \binom{5}{1}y^2 + \binom{5}{2}y^4 + \cdots$$

so the y^4 term is $\binom{5}{2}y^4 = 10y^4$.

Therefore the x^2y^4 term is $21x^2 \times 10y^4 = 210x^2y^4$, and the coefficient is 210; this is option F.

Question 14

We make a quick sketch of $y = mx$ and $y = x^3$ so that we can identify the area enclosed. Note that if $m \leq 0$, then there is no area enclosed, so we must have $m > 0$.



The first thing we note is that there are *two* regions we need to consider: one with $x > 0$ and one with $x < 0$. By symmetry, their areas are equal, so it suffices to find m such that the area of the region with $x > 0$ is 3.

To find the area enclosed for $x > 0$, we need to find where the line and cubic intersect; this is when $x^3 = mx$, so at $x = 0$ and $x = \pm\sqrt{m}$. As we are interested in the point where $x > 0$, this is at $s = \sqrt{m}$.

The enclosed area can be calculated as the area below the line minus the area below the cubic. So as a single integral, this gives

$$\begin{aligned} \int_0^{\sqrt{m}} mx - x^3 \, dx &= \left[\frac{1}{2}mx^2 - \frac{1}{4}x^4 \right]_0^{\sqrt{m}} \\ &= \frac{1}{2}m^2 - \frac{1}{4}m^2 \\ &= \frac{1}{4}m^2 \end{aligned}$$

(We could also have calculated the area under the line as the area of a triangle.)

Since this area is 3, we have $\frac{1}{4}m^2 = 3$, so $m^2 = 12$ and $m = \sqrt{12} = 2\sqrt{3}$. This is option E.

Question 15

We can start by rewriting $\log_2\left(\frac{1}{x}\right)$ as $-\log_2 x$, so the equation simplifies to

$$(\log_2 x)^4 + 12(-\log_2 x)^2 - 2^6 = 0,$$

which further simplifies to

$$(\log_2 x)^4 + 12(\log_2 x)^2 - 2^6 = 0.$$

Writing $y = (\log_2 x)^2$ gives the quadratic

$$y^2 + 12y - 64 = 0.$$

This factorises as

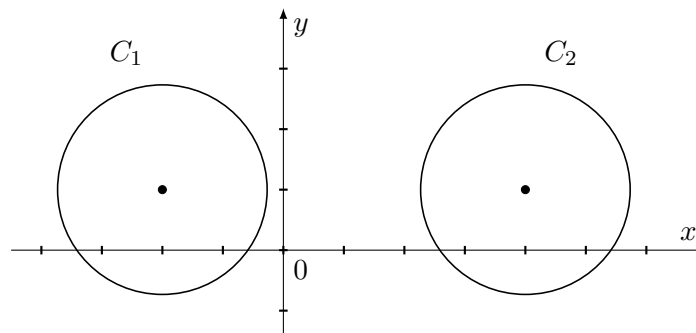
$$(y + 16)(y - 4) = 0,$$

so $y = (\log_2 x)^2 = 4$ (as this cannot be negative).

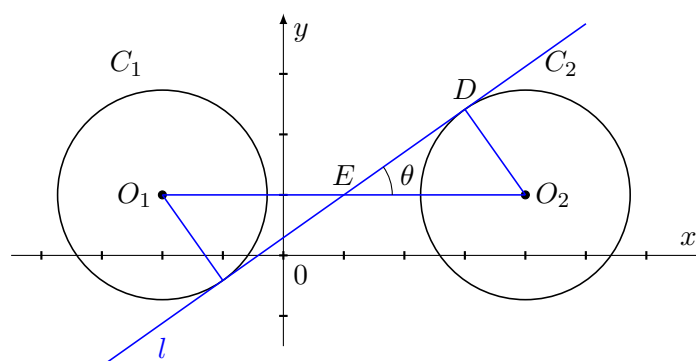
Thus $\log_2 x = 2$ or $\log_2 x = -2$, giving $x = 2^2 = 4$ or $x = 2^{-2} = \frac{1}{4}$. The positive difference between these is $4 - \frac{1}{4} = \frac{15}{4}$, which is option C.

Question 16

It is helpful to sketch the situation before going any further. The circle C_1 has centre $(-2, 1)$ and radius $\sqrt{3}$, while circle C_2 has centre $(4, 1)$ and radius $\sqrt{3}$. A quick sketch shows the configuration:



There are four straight lines which are tangent to both of these circles: two of these are horizontal, and the other two are not. We want the one with a positive gradient, so we sketch it on the graph along with the radii which meet it.



By joining the two circle centres, we see that the angle the line l makes with the x -axis, which is θ , is the same as the angle marked on this diagram. We have also labelled some of the points so that we can refer to them later.

Since the radius at D meets the tangent l at right angles, we can work out the side lengths of the triangle O_2DE and hence $\tan \theta$. The distance between the circle centres is $4 - (-2) = 6$, so the hypotenuse EO_2 has length 3. The circle C_2 has radius $O_2D = \sqrt{3}$, so by Pythagoras's Theorem, $DE^2 = (EO_2)^2 - (DO_2)^2 = 9 - 3 = 6$. Thus $DE = \sqrt{6}$ and hence

$$\tan \theta = \frac{\sqrt{3}}{\sqrt{6}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

and the correct answer is option C.

Question 17

A graph-based (exam-suitable) approach is as follows: Thinking about a sketch graph of $y = \sqrt{x}$, any negative value of m will lead to a single point of intersection, so we must be restricted to $m > 0$. This excludes options D and F. When $c \leq 0$, there can be at most one point of intersection. The only option which rejects all possible values of m when $c < 0$ is A, so that must be the correct option. (This option does not make sense when $c = 0$, but we ignore that.)

An algebraic solution is as follows.

The line and the graph intersect when $mx + c = \sqrt{x}$, where $x \geq 0$. If $m = 0$, there is at most one solution, so we cannot have $m = 0$. The equation rearranges to give a quadratic in $\sqrt{x}$, namely $m\sqrt{x} - \sqrt{x} + c = 0$. For there to be two distinct non-negative solutions, we first require the discriminant to be positive, so $1 - 4mc > 0$, or $4mc < 1$. We can write the roots explicitly as

$$\sqrt{x} = \frac{1 \pm \sqrt{1 - 4mc}}{2m}$$

and we require both of these to be non-negative. The midpoint of the two roots is $\frac{1}{2m}$, so if $m < 0$, one or both of the roots is negative; we must therefore exclude this case. So we must have $m > 0$ and $4mc < 1$. If $c < 0$, then $1 - 4mc > 1$, so $\frac{1 - \sqrt{1 - 4mc}}{2m} < 0$; this therefore gives only one non-negative root.

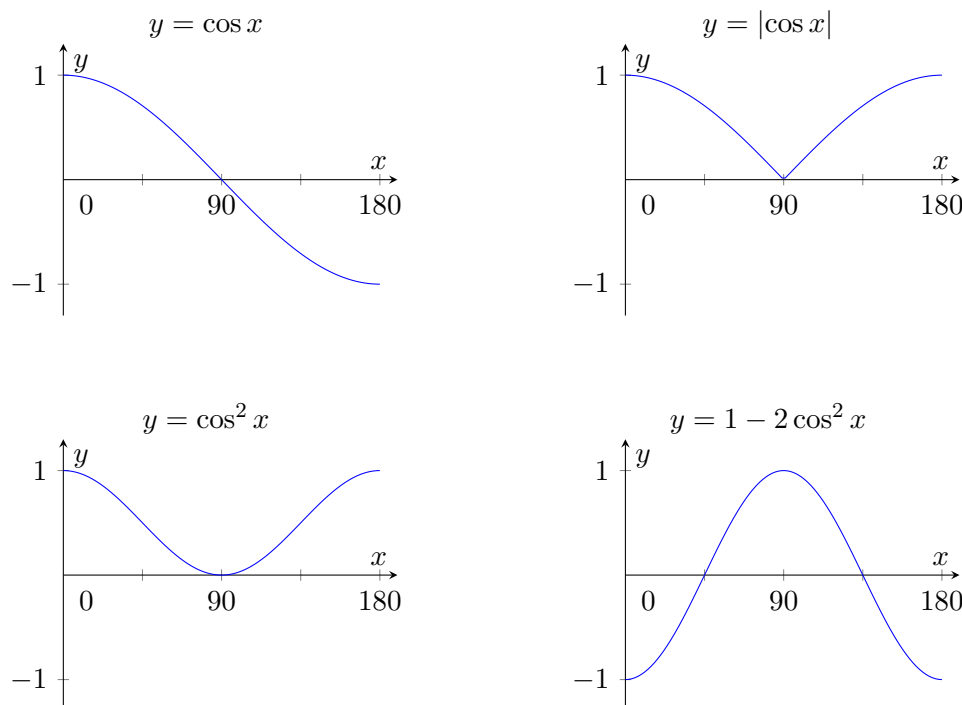
We are thus left with two possibilities: $m > 0$ and $c = 0$, or $m > 0$ and $4mc < 1$, which rearranges as $0 < m < \frac{1}{4c}$. The closest offered option is A.

Question 18

Approach 1: graph sketching

One way to start this question is to sketch the two graphs $y = 1 - 2\cos^2 x$ and $y = |\cos x|$. It may not be obvious how to sketch the first of these graphs (unless you happen to know about double-angle formulae), but we can sketch something adequate for this question.

We can sketch the graph of $y = \cos x$, then use that to sketch both $y = |\cos x|$ and $y = \cos^2 x$, and then the latter one can be used to sketch $y = 1 - 2\cos^2 x$. Squaring $\cos x$ gives a graph which is never negative, and is zero when $\cos x = 0$. So the graphs look like this:



Imagining overlaying the graphs of $y = |\cos x|$ and $y = 1 - 2\cos^2 x$, we see that there are two points of intersection symmetrically placed around 90° , so their sum is 180° , which is option A.

Approach 2: using algebra

We solve the two equations $1 - 2\cos^2 x = \cos x$ when $\cos x \geq 0$ and $1 - 2\cos^2 x = -\cos x$ when $\cos x < 0$.

First, $1 - 2\cos^2 x = \cos x$ when $\cos x \geq 0$ gives $2\cos^2 x + \cos x - 1 = 0$, which factorises as $(2\cos x - 1)(\cos x + 1) = 0$. This has solutions $\cos x = \frac{1}{2}$ and $\cos x = -1$. We must reject the second solution as we require $\cos x \geq 0$, so we have the solution $x = 60^\circ$ (in the range $0^\circ \leq x \leq 180^\circ$).

Next, $1 - 2\cos^2 x = -\cos x$ when $\cos x < 0$ gives $2\cos^2 x - \cos x - 1 = 0$, which factorises as $(2\cos x + 1)(\cos x - 1) = 0$. This has solutions $\cos x = -\frac{1}{2}$ and $\cos x = 1$, of which only the first is valid. This gives the solution $x = 120^\circ$.

Therefore there are two solutions and their sum is $60^\circ + 120^\circ = 180^\circ$.

Question 19

Let us write $f(x) = x^2 - 52x - 52$. We first note that $f(x)$ is a quadratic, and that $f(0) = -52$. We might notice that $f(52) = 52^2 - 52^2 - 52 = -52$, or we could complete the square to find that $f(x) = (x - 26)^2 - 26^2 - 52$, so the line of symmetry is at $x = 26$, again giving $f(52) = -52$.

It follows that $f(x) < 0$ when $0 \leq x \leq 52$ (and other values of x just outside this range).

We next check $f(53)$. We have $f(53) = 53^2 - 52 \times 53 - 52 = 53^2 - 52 \times 54$. Now $52 \times 54 = (53 - 1)(53 + 1) = 53^2 - 1$, so $f(53) = 53^2 - (53^2 - 1) = 1$.

Therefore 53 is the smallest integer for which $f(x) > 0$, and the correct answer is option E.

Question 20

The equation has solutions $x = a$ and $x = \frac{1 \pm \sqrt{1-4a}}{2}$.

One case in which there are two distinct values is when $1 - 4a = 0$ and $a \neq \frac{1}{2}$, which occurs when $a = \frac{1}{4}$.

The other case in which there are two distinct values is when a is one of the solutions of the quadratic $x^2 - x + a = 0$. We could find this by equating $a = \frac{1 \pm \sqrt{1-4a}}{2}$, but a much simpler approach is to substitute $x = a$ into the quadratic, giving $a^2 - a + a = 0$, hence $a^2 = 0$ giving $a = 0$.

Therefore there are two such values, $a = 0$ and $a = \frac{1}{4}$, which is option C.

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This document contains maths equations and symbols which at present we are not able to produce adequately for accessibility. We are working to improve this.

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Test of Mathematics for University Admission, 2020 Paper 2

Worked Solutions

Version 1.0, April 2021

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Introduction for students

These solutions are designed to support you as you prepare to take the Test of Mathematics for University Admission. They are intended to help you understand how to answer the questions, and therefore you are strongly encouraged to **attempt the questions first** before looking at these worked solutions. For this reason, each solution starts on a new page, so that you can avoid looking ahead.

The solutions contain much more detail and explanation than you would need to write in the test itself – after all, the test is multiple choice, so no written solutions are needed, and you may be very fluent at some of the steps spelled out here. Nevertheless, doing too much in your head might lead to making unnecessary mistakes, so a healthy balance is a good target!

There may be alternative ways to correctly answer these questions; these are not meant to be ‘definitive’ solutions.

The questions themselves are available on the ‘**Preparing for the test**’ section on the Admissions Testing website.

Question 1

We find the points where the line and curve meet by solving the equations simultaneously; this gives $x^2 + kx + 2 = x - 2$. Rearranging gives

$$x^2 + (k - 1)x + 4 = 0.$$

The discriminant of this is $(k - 1)^2 - 4 \times 1 \times 4 = (k - 1)^2 - 16$. The line and curve cross or touch when the discriminant is ≥ 0 , so we must solve

$$(k - 1)^2 - 16 \geq 0.$$

Expanding and simplifying this gives

$$k^2 - 2k - 15 \geq 0,$$

which we can factorise to get

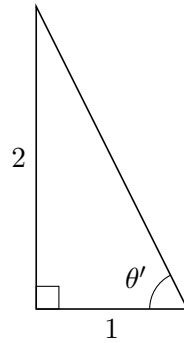
$$(k - 5)(k + 3) \geq 0.$$

The equation $(k - 5)(k + 3) = 0$ has roots $k = -3$ and $k = 5$, so the quadratic is non-negative when $k \leq -3$ or $k \geq 5$. Thus the correct solution is E.

Question 2

In the range $180^\circ < \theta < 360^\circ$, we have that $\tan \theta > 0$ when $180^\circ < \theta < 270^\circ$. In this smaller range, $\cos \theta < 0$.

Next, in the range $0^\circ < \theta' < 90^\circ$, if $\tan \theta' = 2$, we can sketch a right-angled triangle with this angle:



The hypotenuse has length $\sqrt{2^2 + 1^2} = \sqrt{5}$, so $\cos \theta' = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}$.

Therefore for our original angle θ , we have $\cos \theta = -\frac{\sqrt{5}}{5}$, which is option F.

Question 3

Lines (I) to (V) are algebraic manipulations; we need to check whether each manipulation is correct.

Line (I) writes $4 = 2 \times 2$, which is correct.

Line (II) multiplies out the inner brackets, but does not do it correctly: the inner brackets should expand to $(9n + 1) - (3n - 1)$, which simplifies to $9n + 1 - 3n + 1$.

The remaining manipulations are correct.

Thus the first error occurs on line (II), and the correct option is C.

Question 4

A counterexample is a positive integer N that is greater than 6 but it is not the case that N can be written as the sum of two non-prime integers that are greater than 1.

- $N = 5$ is not greater than 6, so is not a counterexample.
- $N = 7 = 2 + 5 = 3 + 4$ cannot be written in this way, so this provides a counterexample.
- $N = 9 = 2 + 7 = 3 + 6 = 4 + 5$ also cannot be written in this way.

Therefore both II and III provide counterexamples; the correct option is G.

Question 5

Note that none of these graphs have scales on.

A key question is what happens to y when x is large and positive and when x is large and negative.

If $x = 100$, say, then $y = \frac{2^{100}}{1 + 2^{100}}$. Since 2^{100} is a very large positive number, this fraction is almost exactly 1. This excludes graphs B and C, where the value of y increases towards infinity. It also excludes graphs D and E, where the value of y tends towards 0. We are therefore left with graphs A and F.

One way to distinguish between A and F is to compare the value at $x = 0$ with the value when x is large. When $x = 0$, we have $y = \frac{1}{2}$. Since $\frac{1}{2} < 1$ and y is close to 1 when x is large, graph F is not possible, so the answer is graph A.

Alternatively, we can consider what happens when $x = -100$, say. Then $y = \frac{2^{-100}}{1 + 2^{-100}}$. Since 2^{-100} is a tiny positive number, very close to 0, this fraction is almost exactly 0. This excludes all of the graphs except for A and B; combining this with our earlier observations shows that the correct graph is A.

Question 6

If we think about the integrals as meaning ‘the (signed) area under the graph of $y = f(x)$ between $x = -5$ and $x = 0$ ’ and the same between $x = 0$ and $x = 5$, we realise that none of these conditions is necessary (though the first two conditions are actually sufficient).

To show fully that a condition is not necessary, we need to find an example of a function for which the integral equation is satisfied but for which the condition is not.

An example of a function which satisfies the integral equation but does not satisfy conditions II or III is the function $f(x) = x^2$.

An example of a function which satisfies the integral condition but does not satisfy condition I is $f(x) = \sin 2\pi x$, which has 5 full periods in each of the ranges $-5 \leq x \leq 0$ and $0 \leq x \leq 5$, so each of the integrals is equal to 0, but it is not the case that $f(x) = f(-x)$ for $-5 \leq x \leq 5$.

Therefore the correct option is A.

Question 7

To show that a condition is sufficient, we must explain why having the condition means that the parallelogram must be a square. To show that a condition is not sufficient, we must find an example of a parallelogram that satisfies the condition but is not a square.

For condition I, if $PQRS$ is a rhombus that is not a square, then it satisfies the condition but is not a square. So condition I is not sufficient.

For condition II, the same example works: the diagonals of a rhombus intersect at right angles even when the rhombus is not a square.

For condition III, we know that the adjacent angles $\angle PQR$ and $\angle QRS$ sum to 180° , so if condition III is satisfied, they must each be 90° , a right angle. But a rectangle that is not a square satisfies this condition and is a parallelogram.

Therefore none of these conditions is sufficient, and the correct option is H.

Question 8

To prove that a ‘for all’ statement is true, it is necessary to show that the formula holds for every choice of the variables. On the other hand, to show that it is false, it is sufficient to just find a single counterexample, because that shows that it is not true for *all* choices of the variables.

This student has provided a correct counterexample (and it is easy to check that their arithmetic is correct). Therefore the student has correctly shown that (*) is false, and the correct option is E.

One can also observe that ‘for all real numbers a and b , $|a + b| \leq |a| + |b|$ ’ is a true statement, but noting that is different from showing that (*) is false.

Question 9

The student wishes to evaluate $f(4)$. Writing 4 rad to explicitly indicate radians, this becomes

$$f(4) = 4 \sin(4 \text{ rad}) = 4 \times \sin(4 \text{ rad}).$$

Now

$$\pi \text{ rad} = 180^\circ,$$

so dividing by π gives

$$1 \text{ rad} = \frac{180^\circ}{\pi}.$$

Multiplying this by 4 then gives

$$4 \text{ rad} = \frac{4 \times 180^\circ}{\pi}.$$

Therefore the student could type into their calculator

$$4 \times \sin(4 \times 180 \div \pi)$$

which is equivalent to the calculation of F .

Question 10

One result about inequalities is that if $p < q$ and $r < s$, then $p + r < q + s$. This is because

$$p + r < q + r < q + s$$

where the first inequality follows by adding r to $p < q$ and the second follows by adding q to $r < s$.

Since we have $a + b < c + d$ and $a + c < b + d$, we can therefore add these to get

$$2a + b + c < b + c + 2d.$$

Subtracting $b + c$ from both sides gives $2a < 2d$, so $a < d$. Hence I must be true.

A similar approach does not seem to work to show $b < c$, which suggests that perhaps we should look for a counterexample. If d is much larger than a , then the values of b and c should not make much of a difference to the inequality if they are both small. So let us try $a = 1$, $d = 100$, $b = 2$, $c = 1$. Then we do indeed have $0 < 1 + 2 < 1 + 100$ and $0 < 1 + 1 < 2 + 100$, so the first two inequalities hold but $b > c$. Therefore II might not be true.

For III, we note that $a + b > 0$ and $c + d > a + b > 0$, so adding $a + b > 0$ and $c + d > 0$ gives $a + b + c + d > 0$, and thus III must be true.

The correct option is F.

Question 11

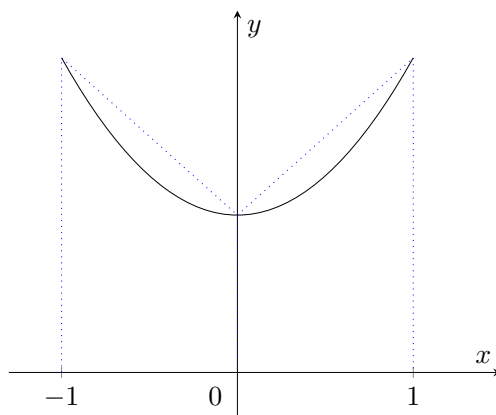
From the nature of the pattern, we see that it sort-of repeats every 2 units along each axis. So let us simplify this question by replacing 99 and 100 by 3 and 4 (an odd number and the even number one more than it).

Looking at the portion of the graph we are given, we see the points $(3, 4)$ and $(4, 3)$ are both on the spiral (top right), $(3, -4)$ and $(4, -3)$ are too (bottom right), $(-4, -3)$ and $(-3, -4)$ are both on the bottom left and if we continue the line at the top, $(-3, 4)$ is at the top left. However $(-4, 3)$ is missed out: that is the gap at the end of one 'loop' as the spiral heads out to form the next 'loop'.

Therefore in the case of 99 and 100, $(-100, 99)$ will be missed, which is option G.

Question 12

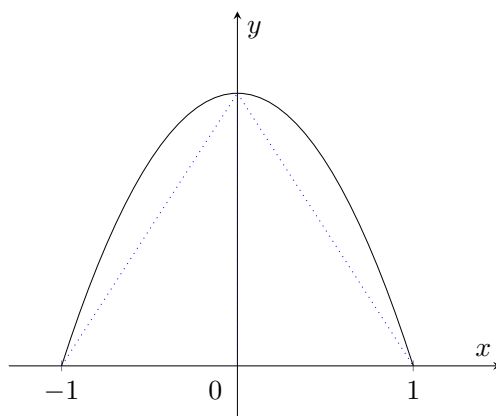
The trapezium rule can produce an overestimate even if the function's gradient goes from negative to positive, for example with the function $f(x) = x^2 + 1$ between $x = -1$ and $x = 1$:



Therefore neither the condition $f'(x) > 0$ for all x with $a < x < b$ nor the condition $f'(x) < 0$ for all x with $a < x < b$ is necessary for the trapezium rule to produce an overestimate. This rejects the 'only if' options (B, C, E, F), leaving us with only A and D.

The above quadratic can also be used to help distinguish between these two options. We have $f'(x) = 2x$ and $f''(x) = 2$. Thus for $-1 < x < 0$, $f'(x) < 0$ and $f''(x) > 0$ and the trapezium rule (with one strip) gives an overestimate. So D appears plausible.

On the other hand, we could reflect the quadratic to change the sign of $f''(x)$: consider the function $f(x) = 1 - x^2$.



We now have $f'(x) = -2x$ and $f''(x) = -2$. So when $-1 < x < 0$, we have $f'(x) > 0$ and $f''(x) < 0$, yet the trapezium rule (with one strip) produces an underestimate. This shows that A cannot be correct.

By elimination, the correct option must be D.

It is also possible to justify this: a function with $f''(x) > 0$ is 'bending upwards' (the technical term is *convex*). Any time we draw a chord between two points on the graph, the chord will lie above the graph (except at the endpoints), so the trapezium rule estimate will be an overestimate of the integral. In fact, this argument makes no assumptions about the sign of either $f(x)$ or of $f'(x)$; having $f''(x) > 0$ for $a < x < b$ is sufficient for the trapezium rule to give an overestimate (but it is not necessary).

Question 13

This question is about inequalities. To begin with, the only thing that we know for certain is that $(f(x))^2 \geq 0$ for all x , as it is the square of a real number. It follows that $\int_0^1 f(x) \, dx \geq \int_0^3 f(x) \, dx$, as we are adding a non-negative number to $\int_0^3 f(x) \, dx$ to get $\int_0^1 f(x) \, dx$. Therefore II is necessarily true.

Let us now take II and rewrite the integral from 0 to 3 as the sum of the integrals from 0 to 1 and from 1 to 3; this is so that we can match the integral on the right hand side. We obtain:

$$\int_0^1 f(x) \, dx + \int_1^3 f(x) \, dx \leq \int_0^1 f(x) \, dx$$

and we can now subtract the integral from 0 to 1 to get

$$\int_1^3 f(x) \, dx \leq 0.$$

Since the integral is negative or zero, the function itself must be so somewhere between 1 and 3; if it were positive throughout the interval, the integral would be positive. Therefore condition I must also be true.

Hence the correct answer is option D.

Question 14

Approach 1: direct reasoning

We can see from the example exactly what is needed for a sequence T to have property P: the sum of the terms from the $(m+1)$ th term to the $(2m)$ th term must be zero.

Consider now the two statements.

- I We will have $ad < 0$ if $a > 0$ and $d < 0$. But if $|d|$ is large enough, then the second term will be negative and there is no hope of a sum of terms giving zero. For example, the sequence $1, -2, -5, \dots$ has $ad = 1(-2) = -2 < 0$, but does not have property P. So this statement is false.
- II In the given example, d is even and m is also even. But if m is odd, we would need the middle term of the second half to be zero, and then it does not matter whether d is even or odd, for example in the sequence (and this was constructed by working backwards): $4, 3, 2, 1, 0, -1, \dots$, which has property P taking $m = 3$. An even more extreme example is the sequence $1, 0, \dots$, where $d = -1$ again and $m = 1$. Therefore this statement is false.

Hence the correct option is A.

Approach 2: algebraic approach

The sum of the first m and the first $2m$ terms of the sequence are given by

$$\begin{aligned} S_m &= \frac{1}{2}m(2a + (m-1)d) \\ S_{2m} &= \frac{1}{2}(2m)(2a + (2m-1)d) \end{aligned}$$

and these are equal if and only if

$$\frac{1}{2}m(2a + (m-1)d) = \frac{1}{2}(2m)(2a + (2m-1)d).$$

Dividing both sides by $\frac{1}{2}m$ gives

$$2a + (m-1)d = 2(2a + (2m-1)d).$$

Expanding the brackets and rearranging gives

$$2a + (3m-1)d = 0. \quad (*)$$

- I The equation $(*)$ can be rearranged to make m the subject, giving

$$m = \frac{1}{3} \left(-\frac{2a}{d} + 1 \right).$$

There is no reason why this should be an integer whenever $ad < 0$. For example, taking $a = 1$ and $d = -5$ gives $m = \frac{1}{3} \times \frac{7}{5}$ which is not an integer. Therefore this statement is false.

- II We can rearrange $(*)$ to make d the subject:

$$d = -\frac{2a}{3m-1}$$

but it is actually more helpful to rearrange it as

$$3m - 1 = -\frac{2a}{d}.$$

To show that the statement is true, we need to show that whenever we have integer values for a , d and m , d will be even; to show that the statement is false, we need to find an example where d is odd. If we take $d = 1$, we need a value of a for which $-2a$ is one less than a (positive) multiple of 3; $-2a = 2$ and $-2a = 8$ both work, giving $a = -1$ and $a = -4$ respectively. Similarly, if we take $d = 3$, we need a value of a for which $-\frac{2a}{3}$ is one less than a (positive) multiple of 3; here, $-2a = 2 \times 3 = 6$ would work, giving $a = -3$.

Any one of these counterexamples would show that this statement is false.

Again, we see that the correct option is A.

Question 15

It is very helpful to write out the values of the summands (the terms being added) in terms of $\sqrt{3}$ and similar, and then work out the first few values of the sum.

k	1	2	3	4	5	6	7	8	9	10	...
$\frac{k\pi}{3}$	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π	$\frac{7\pi}{3}$	$\frac{8\pi}{3}$	3π	$\frac{10\pi}{3}$	...
$\sin(\frac{k\pi}{3})$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{2}$	0	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{2}$	0	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{2}$	0	$-\frac{\sqrt{3}}{2}$	...
$\sum_{k=1}^n \sin(\frac{k\pi}{3})$	$\frac{\sqrt{3}}{2}$	$2\frac{\sqrt{3}}{2}$	$2\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{2}$	0	0	$\frac{\sqrt{3}}{2}$	$2\frac{\sqrt{3}}{2}$	$2\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{2}$	...

We can see that the sum repeats every 6 steps, and that it equals $\frac{\sqrt{3}}{2}$ twice in every cycle of 6, specifically when $n = 1, 4, 7, 10, \dots$. This sequence of n values are the numbers which are 1 more than a multiple of 3, and so the correct option is D.

Question 16

We can check whether the calculation stands any chance of being correct by checking whether line (V) is correct. We can work out the integral directly:

$$\begin{aligned}\frac{d}{dx} \left(\int_x^{2x} t^2 dt \right) &= \frac{d}{dx} \left(\left[\frac{1}{3} t^3 \right]_x^{2x} \right) \\ &= \frac{d}{dx} \left(\frac{1}{3} (2x)^3 - \frac{1}{3} x^3 \right) \\ &= \frac{d}{dx} \left(\frac{7}{3} x^3 \right) \\ &= 7x^2\end{aligned}$$

So an error has certainly been introduced somewhere.

From this, it is clear that line (IV) is also incorrect.

Line (I) is correct, as $\int_a^b f(t) dt + \int_b^c f(t) dt = \int_a^c f(t) dt$ for all functions $f(t)$ and real numbers a , b and c .

Therefore the error must lie in line (II) or (III). We can work out the left hand sides of the equations explicitly as we did for line (V):

$$\begin{aligned}\frac{d}{dx} \left(\int_0^x t^2 dt \right) &= \frac{d}{dx} \left(\left[\frac{1}{3} t^3 \right]_0^x \right) \\ &= \frac{d}{dx} \left(\frac{1}{3} x^3 \right) \\ &= x^2\end{aligned}$$

so line (II) is correct. However,

$$\begin{aligned}\frac{d}{dx} \left(\int_0^{2x} t^2 dt \right) &= \frac{d}{dx} \left(\left[\frac{1}{3} t^3 \right]_0^{2x} \right) \\ &= \frac{d}{dx} \left(\frac{8}{3} x^3 \right) \\ &= 8x^2\end{aligned}$$

so line (III) is incorrect. Hence the correct option is D.

The reason for the error is that the fundamental theorem of calculus is stated for the case where the upper limit is x ; if the upper limit is $2x$, then the integral grows twice as fast as x increases, so we actually have $\frac{d}{dx} \left(\int_0^{2x} t^2 dt \right) = 2(2x)^2 = 8x^2$.

Question 17

Let us write the six integers as two sets of three, each set in increasing order:

$$a, 8, b \quad c, 9, d$$

We know the middle integers are 8 and 9 because these are the medians.

The first set of three has a mean of 10 and so they must sum to $3 \times 10 = 30$, therefore $a + b = 22$. Likewise, the second set sums to 36 and so $c + d = 27$.

We also know that $a \leq 8 \leq b$ and $c \leq 9 \leq d$.

Here are tables of some of the possibilities for a and b and for c and d ; the two tables are independent of each other:

a	b	c	d
8	14	9	18
7	15	8	19
6	16	7	20
5	17	6	21
4	18	5	22
3	19	4	23

The range of the set of all six integers is the larger of b and d minus the smaller of a and c .

As all six integers are distinct, we cannot have $a = 8$ or $c = 9$ or $c = 8$. We also want b and d to be as small as possible while a and c are as large as possible to get a small range.

Let us try $a = 7$, $b = 15$. Then we cannot have $c = 7$, so we try $c = 6$, $d = 21$. This gives a range of $21 - 6 = 15$.

If we now try $a = 6$, $b = 16$, we can take $c = 7$, $d = 20$, and this gives a range of $20 - 6 = 14$, which is smaller.

If we next try $a = 5$, $b = 17$, we can again take $c = 7$, $d = 20$, but now the range has grown to $20 - 5 = 15$ again. And if we go smaller with the choice of a , say $a = 4$, $b = 18$, then the smallest range we can obtain becomes even larger: $20 - 4 = 16$. The range will keep growing as we take smaller values for a .

So the smallest possible range is 14, which is option E.

Question 18

Let us write $f(x) - g(x)$ explicitly:

$$f(x) - g(x) = (ax^3 + bx^2 + cx + d) - (px^3 + qx^2 + rx + s) = (a - p)x^3 + (b - q)x^2 + (c - r)x + (d - s).$$

We are given that this cubic is positive for every real x . But a cubic always crosses the x -axis! The only way to reconcile this is if the cubic $f(x) - g(x)$ is not actually a cubic, because $a - p = 0$.

So we have $a - p = 0$ and statement I is false. We can also now write

$$f(x) - g(x) = (b - q)x^2 + (c - r)x + (d - s).$$

If this function is always positive, then there are two possibilities: either $b - q$ is nonzero and the quadratic has a negative discriminant, or $b - q = 0$ and this is not actually a quadratic. The second option looks quite like statement II: if $b - q = 0$, that is, $b = q$, then we have

$$f(x) - g(x) = (c - r)x + (d - s).$$

Since a linear function always crosses the x -axis, this cannot be always positive unless $c - r = 0$, in which case the function is actually just the constant $f(x) - g(x) = (d - s)$. (To be more precise, this is still a linear function, just with zero x coefficient.) Therefore, if $f(x) - g(x) > 0$ for all real x , it follows that if $b = q$, then $c = r$, and so statement II is true.

We have reached the end of this line of reasoning, and we have not stumbled upon anything that looks like statement III. But d and s are the constant coefficients in the polynomials f and g , so if we take $x = 0$, we have $f(0) = d$ and $g(0) = s$. Since $f(0) - g(0) > 0$, this shows that $d - s > 0$, or $d > s$, hence statement III is true.

The correct option is G.

Question 19

The person in the centre gives us a lot of information. If they are a truth-teller, then all four neighbours are liars, but if they are a liar, then at least one neighbour is a truth-teller. Since we are only interested in the number of truth-tellers and not their positions, we can assume the person directly ‘above’ them is a truth-teller. So we have these two possibilities (where T means truth-teller and L means liar):

	L	
L	T	L
	L	

	T	
	L	

In the first possibility, each corner position has two neighbours, both of whom are liars, so they must be occupied by truth-tellers. In the second possibility, the top two corners must be liars, giving these possibilities:

T	L	T
L	T	L
T	L	T

L	T	L
	L	

Case 1: 5 T, 4 L

In the right-hand possibility, the corner liars already have one neighbour who is a truth-teller, so the other neighbour could be either a truth-teller or a liar. This gives four possibilities for the left side and right side people: both liars, both truth-tellers or one of each in either order. By reflecting the grid, we only need to consider one case of one liar, one truth-teller, so we now have these three possible ways of continuing the grid:

L	T	L
L	L	L

L	T	L
T	L	T

L	T	L
L	L	T

Every truth-teller is surrounded by liars, and every liar must have at least one truth-teller as a neighbour, so we can fill in some of the bottom row in each of these three possibilities:

L	T	L
L	L	L
T		T

L	T	L
T	L	T
L		L

L	T	L
L	L	T
T		L

We can now fill in the final square in each of these, and check that the three people in the bottom row are consistent with the rules:

L	T	L
L	L	L
T	L	T

Case 2: 3 T, 6 L

L	T	L
T	L	T
L	T	L

Case 3: 4 T, 5 L

L	T	L
L	L	T
T	L	L

Case 4: 3 T, 6 L

We therefore have four cases. The smallest number of truth-tellers is in cases 2 and 4, with 3 truth-tellers, and the largest number is in case 1 with 5 truth-tellers. Therefore the correct option is E.

Question 20

It is helpful to write all six statements as ‘if ... then’ statements:

- A **if** $x \geq 0$ **then** $f(x) < 0$
- B **if** $f(x) \geq 0$ **then** $x < 0$
- C **if** $x \geq 0$ **then** $f(x) \geq 0$
- D **if** $x < 0$ **then** $f(x) < 0$
- E **if** $f(x) \geq 0$ **then** $x \geq 0$
- F **if** $x < 0$ **then** $f(x) \geq 0$ **and if** $f(x) \geq 0$ **then** $x < 0$

If F is true, then B is also true, so F is false.

Let us suppose that A is true. Then its contrapositive is also true. The contrapositive of A is: **if** $f(x) \geq 0$ **then** $x < 0$, which is statement B. So A and B are either both true or both false. Since exactly one statement is true, A and B must both be false.

Similarly, D and E are contrapositives of each other, so they must both be false as well.

The only statement left is C, so that must be true.

Furthermore, C contradicts A, so A (and B) must be false; C is independent of D (knowing that C is true does not give any information about whether D is true or false), and C contradicts F, so F must be false.

Therefore having C true and the other five statements false is consistent; the function $f(x) = |x|$ is an example of a function for which C is true and the other five statements are false.

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